STUFF+ ANALYSIS IN THE PROSPECT LEAGUE - 2026

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NORMAL CORNBELTERS

INTRODUCTION

Stuff plus measures how effective a pitch is based on only its physical characteristics. This metric has become more popular in scouting and player analyses within the last decade or so. When evaluating pitchers, it's more beneficial to look at what he can control(how his pitches move) rather than results based metrics where the hitter and defense come into play. This is because when the ball is put in play, the skill level of the defensive players becomes an influential part of what happens on the play. You would much rather look at the physical characteristics of a pitcher's pitch mix. I created this model to accurately predict how good the pitchers in the Prospect League were physically within the 2026 summer season. My goal was to measure future value of pitchers based on their true stuff and not have the impact of the batter or defense included.

DATA ACQUISITION:

All data for this model was taken from Trackman and PrestoSports. Trackman is a 3D Doppler radar used to measure player and ball movement. This system is able to track every movement from the pitcher's hand to Homeplate or off the bat. It can also read all defensive position locations to determine shifts or trends. I used trackman for all pitch metrics such as spin rate, velocity, spin axis, release height, induced vertical break, horizontal break and tagged pitch type. I used PrestoSports for all play-by-play data to complete the run-value matrix and train the stuff+ model. PrestoSports was mainly helpful for the baserunning information. Trackman does not have the ability to track who's on base and what the full situation was, so to get all data I needed, I joined the Trackman file with the box scores from every game in the Prospect league from PrestoSports.

This model uses Run Value as the target variable to train the model. Due to time constraints, I used the MLB run value metrics from FanGraphs. If I were to advance this model even further, I would create a run value matrix for the Prospect League specifically to better tailor the analysis. One thing to note in

this analysis is run values for college baseball tend to by significantly higher than MLB run values. Run values are used to determine the change in run expectancy from one pitch to the next. With the data I had available to me, I started by assigning run values to each play that happened during the game. Once I joined the Trackman data with the PrestoSports box scores, I was able to break it down pitch by pitch and determine the run expectancy per pitch.

METRICS TO KNOW:

Release Speed (RelSpeed): Speed of the ball as it leaves the pitcher's hand.

Release Height (RelHeight): Vertical distance of the ball above home plate when the pitcher releases the ball.

Spin Rate: Speed at which the ball is spinning.

Spin Axis: Direction in which the ball is spinning.

Induced Vertical Break: Distance between the height at which the ball crosses home plate and the height at which it would have crossed if it travelled in a straight line from release and were affected by gravity.

Horizontal Break: Horizontal distance between where the ball crosses and where it would have crossed if it had traveled in a straight line from release and were unaffected by gravity.

**One thing to note about Horizontal break is a negative break is towards first base with positive being toward third base. The metric is taken from a pitcher's perspective looking at home plate rather than a catcher's perspective.

MODELING:

For this model, I decided to use the approach of a classification-based model that uses XGBoost to predict the probability based on pitch outcome classes (ball, called strike, etc.). This helped get the predicted run value per pitch, which was then used for the tagged pitch type.

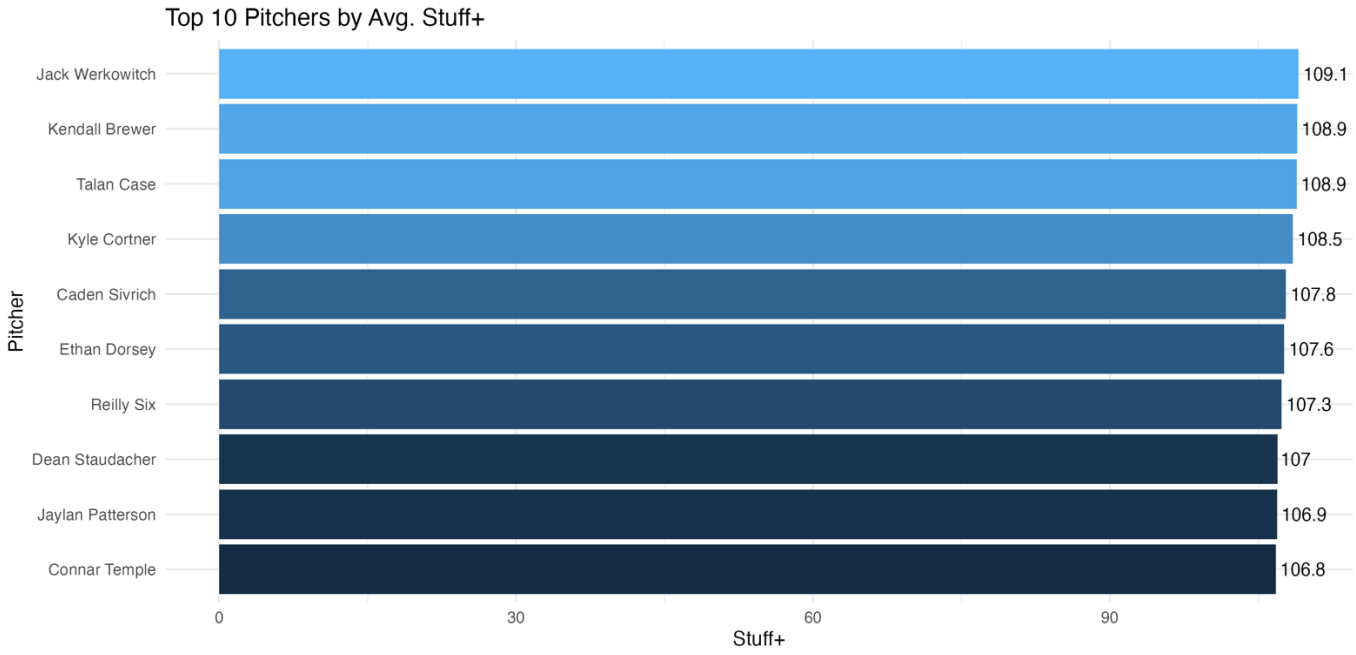
To create the stuff+ variable, I used a formula found in an article on medium.com. The minus sign flips the model so that the players with lower run values have a higher stuff+, meaning the lower the run value is, the more effective the pitch is.

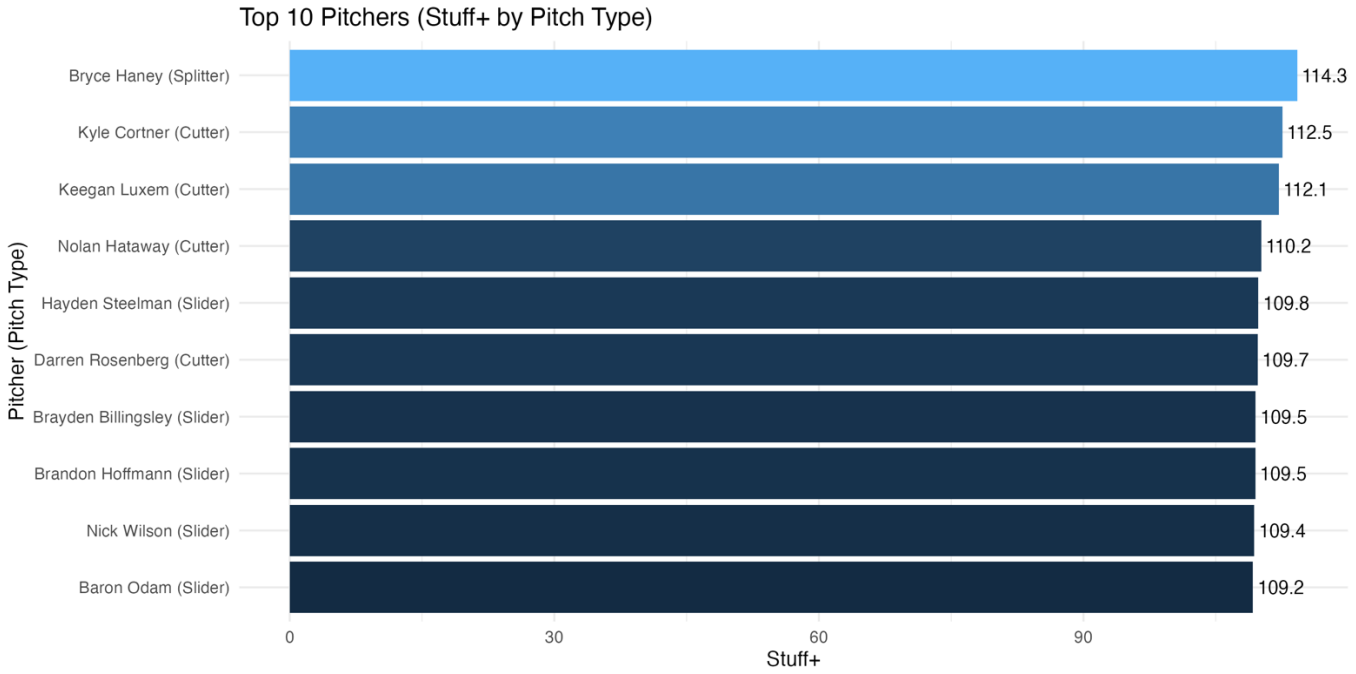
Stuff+ = 100 — [(PRV — μ) / σ] × 10

EVALUATION:

Shown below are three visualizations that show top ten average stuff+ leaders, top ten stuff+ leaders by pitch, and the average stuff+ per pitch type. I chose these visualizations to show that breaking pitches generally perform better in terms of stuff plus, mainly because of the spin. Pitches with the most spin are deemed more effective because the more spin a ball has, the more it will break and become harder for the hitters to recognize and hit. The visualizations show that some pitchers may have one nasty pitch, but their arsenal as a whole is not the most effective. Only three of the top ten pitchers with the most effective pitch made the top ten list of average stuff+ across their entire arsenal.

TaggedPitchType	Avg. Stuff+	Std. Dev. Stuff+
Changeup	94.58	7.07
Curveball	101.09	8.6
Cutter	107.76	5.67
Fastball	99.17	10.94
Sinker	94.72	7.76
Slider	108.88	1.03
Sweeper	117.77	9.37


Top 10 Pitchers by Avg. Stuff+
Jack Werkowitch
Kendall Brewer
Talan Case
Kyle Cortner
Caden Sivrich
Ethan Dorsey
Reilly Six
Dean Staudacher
Jaylan Patterson
Connar Temple
109.1
108.9
108.9
108.5
107.8
107.6
107.3
107
106.9
106.8
Pitcher
0
30
60
90
Stuff+


Top 10 Pitchers (Stuff+ by Pitch Type)
Bryce Haney (Splitter)
Kyle Cortner (Cutter)
Keegan Luxem (Cutter)
Nolan Hataway (Cutter)
Hayden Steelman (Slider)
Darren Rosenberg (Cutter)
Brayden Billingsley (Slider)
Brandon Hoffmann (Slider)
Nick Wilson (Slider)
Baron Odam (Slider)
114.3
112.5
112.1
110.2
109.8
109.7
109.5
109.5
109.4
109.2
Pitcher (Pitch Type)
0
30
60
90
Stuff+

LIMITATIONS:

The main limitation I came across is that this model only looks at the physical characteristics of pitches. There is no consideration of whether the pitcher hit his spot(command) or what the result of the play actually was. With that being said, to get the full story you must add onto this analysis and discuss Command+ and Location+, as well as evaluate the pitchers' results-based stats. Another limitation I found was the lack of data. Trackman was newly implemented league wide this season, so I had one full season of data to work with. Ideally in a predictive model like this, you would have multiple years of data to determine a pitcher's real stuff. The Prospect League is a summer collegiate league, meaning a limitation in the future may be the lack of returning pitchers. Due to most pitchers only playing one season in the Prospect League, the model would only be able to predict future success of a few returning pitchers.

CONCLUSION:

Ultimately this analysis is just a small portion of what goes into evaluating a pitcher's future success. Again, stuff+ only takes the physical characteristics of a pitch into consideration. It would be very insightful to look into location+ as well as pitching+ to show how good the pitchers truly are as a whole. This would consider

whether pitchers hit their spots as well as how nasty their pitches are. That was be the full package of everything in the pitcher's control. Something I found interesting in this analysis was the comparison to the top pitchers in the Prospect League based on ERA, BB's, etc. Almost none of those pitchers were in the top ten of either of the lists I created. This is again because of the limitations of this metric being only the physical characteristics. It is interesting though, to see which pitchers had the "nastiest" pitches in the league and see how they ranked overall based on results.

Overall, I know this is a small part of the overall picture, but I think this is a great start to the full analysis. I'm very proud of the way I was able to apply my skills from my master's coursework as well as learn more coding skills throughout this project. I will now be able to apply all my new skills on future research projects as I continue to grow in my career.